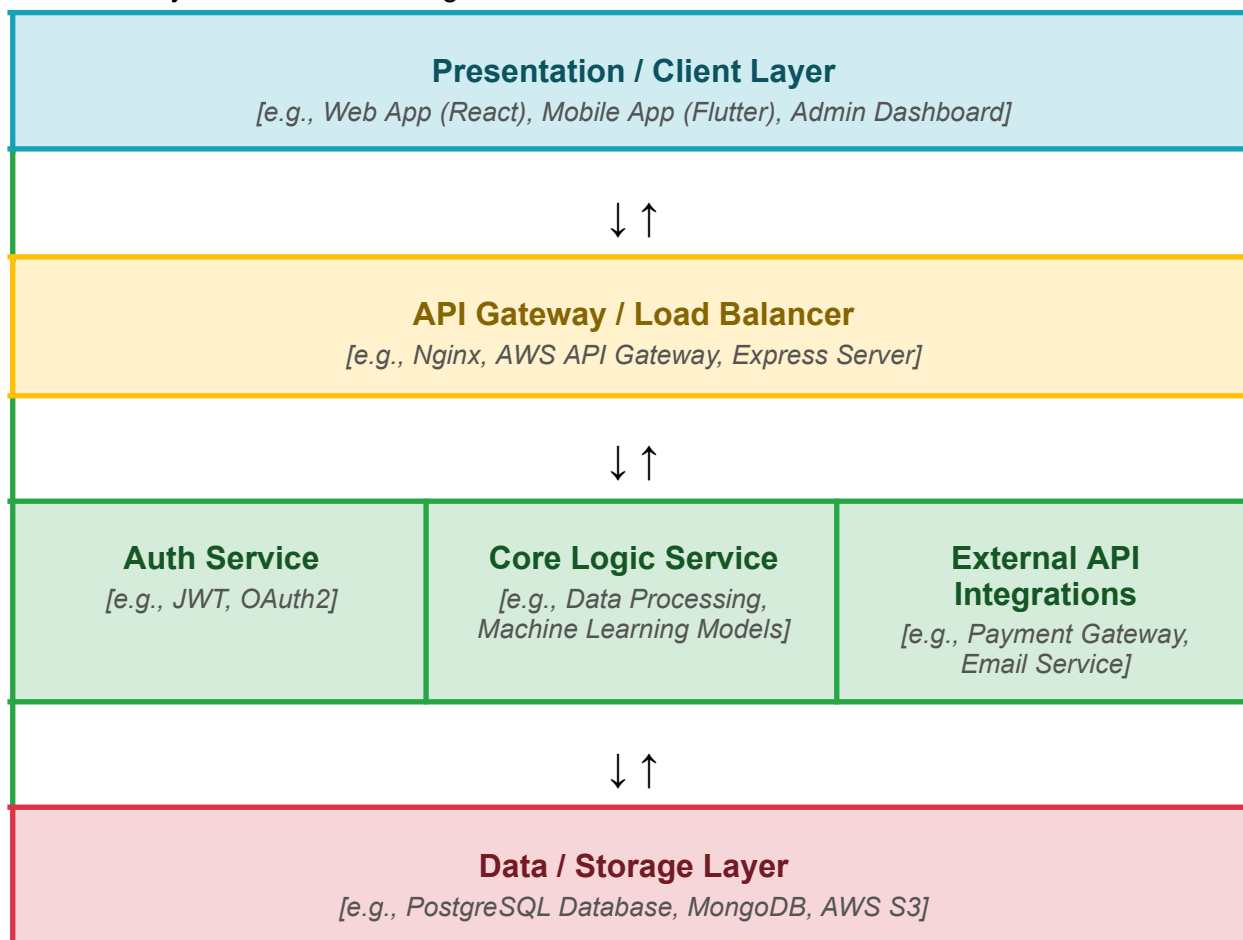


Solution Architecture

Date	26 June 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation layer	Provides responsive SPA with dashboard, loan management, AI negotiation, settlements	React.js,
API Gateway & Middleware	Handles CORS, request routing, and JWT token verification for all protected endpoints	FastAPI, CORSMiddleware,
<i>Financial Engine</i>	Core calculation module: EMI ratio, debt-to-income ratio, stress level classification, health	python-jose, passlib (bcrypt)
	Generates personalized negotiation letters via Google Gemini API with automatic fallback to	google-generativeai (Gemini 1.5 Flash)

Architecture Layer Explanations:

1. Presentation Layer: The responsive user UI built in React.js. It queries backend endpoints and displays clean charts representing monthly expenses, surplus, and loan settlement status.
2. Application Layer: The backend REST API server built in FastAPI. It handles core financial and sorting computations locally and coordinates asynchronous calls to the Google Gemini AI Studio for text generation.
3. Data Layer: The local database system. SQLite manages user credentials, loan listings, and history logs safely, while SQLAlchemy ORM translates python models into standard SQL transactions.